

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1319

Drusilla 100 kV
November 26, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1319. The Interconnection Customer proposes a 141.4 MW solar photovoltaic facility interconnecting at the Drusilla 100 kV substation in the Illinois area. The request was received on October 26, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 480 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,126,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Dooville to Thomas Hill Sta Substation 230kV	Montpelier Electric Generating Station 500-230 kV Autotransformer	101.7	925	5.7
Fairbanks to Biddeford Ip 100kV	Taylorville South 230 kV/1 kV Autotransformer	100.2	347	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dooville to Thomas Hill Sta 230kV	Reconductor line	\$14,075,000	2% (\$234,000)
Fairbanks to Biddeford Ip 100kV	Reconductor line	\$2,239,000	40% (\$892,000)
	Total allocated to project		\$1,126,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.